SC3102: Signals, Systems, and Transforms

Tutorial 03: Z-transform Foundations

1. Answer the following questions on the definition of the Z-transform

(a) For a discrete-time signal, $x[n]$, what is the definition of its Z-transform? Suppose x had Z-transform $X(z)$ with $z = re^{\mathrm{j}\Omega}$ in its ROC, which signal $w[n]$ has its DTFT satisfying:

$$W(\Omega) = X(re^{\mathrm{j}\Omega})$$

(b) Use this result to recover x if $X(z)$ satisfied with $z = 3e^{\mathrm{j}\Omega}$ in its ROC and

$$X(3e^{\mathrm{j}\Omega}) = \frac{1}{1 - \frac{\mathrm{j}}{2}e^{-\mathrm{j}\Omega}}.$$

Hint: First, apply known DTFT inverse pairs (page 3) to recover the re-weighted signal.

(c) For each signal below, find the values of $r > 0$ for which $r^{-n}x[n]$ has a DTFT:

$$x_R[n] = (2\mathrm{j})^n u[n], \qquad x_L[n] = 2^n u[-n-1].$$

(d) Do the following signals have Z-transforms with non-empty ROCs?

$$g[n] = 2^n \qquad h[n] = 2^{|n|}$$

Justify your answer.

2. Consider the following signals:

$$x_1[n] = \delta[n] - 3\delta[n-1] + 2\delta[n-2], \qquad x_2[n] = nu[n],$$
$$x_3[n] = (-2)^n u[-n-1], \qquad x_4[n] = (1/2)^n u[n] - 3^n u[-n-1].$$

(a) For each of these signals, compute its Z-transform and sketch its ROC on the Z-plane.

(b) For each Z-transform, what are the other possible ROCs for the same Z-transform expression?

(c) Which of the above signals have a defined DTFTs? Justify your answer.

3. Invert the following Z-transforms with the indicated methods.

(a) Consider the Z-transform defined below:

$$X(z) = \frac{z^2}{z^2 + 1/4}, \qquad |z| > 1/2.$$

Match the Z-transform to a known pair and compute its inverse.
Hint: Observe the Z-transforms of one-sided sinusoids on page 3.

(b) Consider the following Z-transform expression:

$$A(z) = \frac{3z - 7}{z^2 - 5z + 6}, \qquad |z| > 3.$$

Find the roots p, q of the denominator and compute the partial fraction expansion of A:

$$A(z) = \frac{C}{z - p} + \frac{D}{z - q}.$$

Compute C and D and invert each of the partial fractions according to the supplied ROC.
Hint: To obtain C, multiply both sides by $z - p$ and substitute $z = p$.

(c) For the following Z-transform:

$$Y(z) = \frac{2 + z^{-1}}{1 - \frac{1}{2}z^{-1}}, \qquad |z| > 1/2,$$

use the series expansion method to compute $y[1], y[2],$ and $y[3]$.

(d) Suppose the Z-transform of a signal $v[n]$ was defined as:

$$V(z) = \cdots + \tfrac{1}{8}z^3 + \tfrac{1}{4}z^2 + \tfrac{1}{2}z + 1 + \tfrac{1}{3}z^{-1} + \tfrac{1}{9}z^{-2} + \cdots$$

for z in its ROC. What are $v[-2], v[-1], v[0], v[1], v[2]$?

4. Consider a discrete-time signal $f[n]$ with the following Z-transform expression:

$$F(z) = \frac{2z^2 + 2z}{z^2 + 2z - 8}.$$

(a) Make the pole-zero plot for f and indicate its possible ROCs.

(b) For each $r = 1, 3, 5$, find the ROC and $f[n]$ if the circle $|z| = r$ lies in the ROC.

(c) Suppose $f[n] = 0$ for $n < 0$. Which ROC does it correspond to?

(d) Suppose, instead, that f was two-sided. Which ROC does it correspond to?

(e) Does $|z| = 4$ lie in any of the ROCs? Justify your answer.

Standard Z-transform Pairs and Properties

Signal $x[n]$	Transform $X(z)$	ROC
$\delta[n-k]$	z^{-k}	$0 < \lvert z\rvert < \infty$
$u[n]$	$\dfrac{z}{z-1}$	$\lvert z\rvert > 1$
$-u[-n-1]$	$\dfrac{z}{z-1}$	$0 < \lvert z\rvert < 1$
$p^n u[n]$	$\dfrac{z}{z-p} = \dfrac{1}{1-pz^{-1}}$	$\lvert z\rvert > \lvert p\rvert$
$-p^n u[-n-1]$	$\dfrac{z}{z-p}$	$0 < \lvert z\rvert < \lvert p\rvert$
$p^{n-1} u[n-1]$	$\dfrac{1}{z-p}$	$\lvert z\rvert > \lvert p\rvert$
$-p^{n-1} u[-n]$	$\dfrac{1}{z-p}$	$0 < \lvert z\rvert < \lvert p\rvert$
$np^n u[n]$	$\dfrac{pz}{(z-p)^2}$	$\lvert z\rvert > \lvert p\rvert$
$-np^n u[-n-1]$	$\dfrac{pz}{(z-p)^2}$	$0 < \lvert z\rvert < \lvert p\rvert$
$(n+1)p^n u[n]$	$\dfrac{z^2}{(z-p)^2} = \dfrac{1}{(1-pz^{-1})^2}$	$\lvert z\rvert > \lvert p\rvert$
$(-n-1)p^n u[-n-2]$	$\dfrac{z^2}{(z-p)^2}$	$0 < \lvert z\rvert < \lvert p\rvert$
$\rho^n \cos(\omega_0 n) u[n]$	$\dfrac{z(z-\rho\cos\omega_0)}{z^2 - 2\rho z\cos\omega_0 + \rho^2}$	$\lvert z\rvert > \rho$
$\rho^n \sin(\omega_0 n) u[n]$	$\dfrac{\rho z \sin\omega_0}{z^2 - 2\rho z\cos\omega_0 + \rho^2}$	$\lvert z\rvert > \rho$

Signal	Transform	ROC
$\alpha x[n] + \beta h[n]$	$\alpha X(z) + \beta H(z)$	Contains $\mathcal{R}_X \cap \mathcal{R}_H$.
$x[n-k]$, $k \in \mathbb{Z}$	$z^{-k}X(z)$	Same at finite nonzero z.
$x[-n]$	$X(z^{-1})$	$1/r_2 < \lvert z\rvert < 1/r_1$.
$a^n x[n]$, $a \neq 0$	$X(z/a)$	$\lvert a\rvert r_1 < \lvert z\rvert < \lvert a\rvert r_2$.
$(x * h)[n]$	$X(z)H(z)$	Contains the nonempty common ROC.

DTFT pairs for Question 1. With $W(e^{\mathrm{j}\Omega}) = \sum_n w[n]e^{-\mathrm{j}\Omega n}$ and $0 < |a| < 1$:

Signal $w[n]$	DTFT $W(e^{\mathrm{j}\Omega})$
$a^n u[n]$	$\dfrac{1}{1-ae^{-\mathrm{j}\Omega}}$
$a^{-n} u[-n-1]$	$\dfrac{ae^{\mathrm{j}\Omega}}{1-ae^{\mathrm{j}\Omega}}$